

Executive Summary

Project: Retail Returns & Reverse Logistics Optimization

1. Business Context

The project analyzes retail transaction and cancellation data to identify cancellation patterns, evaluate their potential business and financial impact, and recommend improvements to the retail cancellation and reverse-logistics process.

The project follows a Business Analyst approach combining business requirements, data analysis, reporting, AI-assisted insights, and process improvement.

2. Business Problem

The retailer processes a large volume of online transactions, making it difficult to quickly identify products, markets, time periods, and transactions associated with higher cancellation activity.

The absence of a centralized analytical view creates challenges in monitoring cancellation performance and prioritizing improvement opportunities.

3. Solution

The proposed solution combines:

- Excel for data preparation and exploratory analysis.
- SQL for structured analysis and validation.
- Power BI for interactive KPI monitoring and visualization.
- Generative AI for insight generation and decision support.
- BA techniques including BRD, stakeholder analysis, business rules, gap analysis, user stories, AS-IS/TO-BE process analysis, and implementation planning.

4. Key Analytical Findings

The analysis identified approximately:

- **1M total transactions**
- **19K cancelled transactions**
- **1.83% overall cancellation rate**
- **1.40M cancellation transaction value**

The cancellation rate decreased between the two analyzed periods, indicating an improvement in cancellation performance.

The analysis also identified differences in cancellation activity across products, countries, and time periods.

5. Business Solution

The proposed future-state approach provides:

Transaction Data → Excel/SQL Analysis → Power BI Dashboard → AI-Assisted Insights → Business Validation → Root-Cause Investigation → Improvement Actions → KPI Monitoring

This creates a structured process for moving from raw transaction data to actionable business decisions.

6. Expected Business Benefits

The solution is expected to:

- Improve visibility into cancellation performance.
- Identify high-priority products and markets.
- Improve monitoring of financial impact.
- Support data-driven decision-making.
- Improve data-quality awareness.
- Provide continuous KPI monitoring.
- Reduce the effort required to interpret analytical findings.

7. BA Deliverables

The project demonstrates practical Business Analyst capabilities through:

- Business Requirements Document
- Stakeholder Analysis
- Functional and Non-Functional Requirements
- Business Rules
- Data Requirements
- Gap Analysis
- AS-IS / TO-BE Process Analysis
- User Stories
- Acceptance Criteria
- Requirements Traceability Matrix
- Excel Analysis
- SQL Analysis

- Power BI Dashboard
- AI-Assisted Insights
- Business Recommendations
- Implementation Roadmap

8. Conclusion

The project demonstrates how a Business Analyst can combine business requirements, data analysis, visualization, Generative AI, and process improvement to address a real-world retail business problem.

The solution is designed as a decision-support and process-improvement framework rather than a production system, with recommendations requiring validation by relevant business stakeholders before implementation.